Input shaft is the main shaft of gear box. Output shaft is the propeller shaft. The parts of the overdrive can be seen in Figure 3.8 (b).



Figure 3.8 (b) Parts of overdrive

An epicyclic gear train is mounted on the input shaft. In this epicyclic gear train, a sun gear, three planet gears, a planet carrier and a ring gear are present. This can be seen in the above Figure.

Working Principle

The casing is stationary when the sun gear is locked with casing. Now the overdrive unit starts functioning and the speed of the output shaft is increased. When the sun gear is locked with the planet carrier, the usual input shaft speed is given to the output shaft.

Advantages of overdrive

1. Engine life is increased
2. Achieves reduction in fuel consumption.
3. Less vibration
4. Low noise
5. Excess gear ratio can be achieved
6. Wear of engine is lower.
7. Connecting over drive unit is easy.

Trouble shooting of gear box

1. Gear box noise

Causes	Remedies
1. Bearing may get damaged	Need to change bearing
2. Lubricating oil level may be low	Fill the required amount of lubricating oil
3. Teeth's may be broken in the gear	Need to change the gear
4. Misalignments between gearbox and engine	Need to correct it

2. Hard Gear Shift

Causes	Remedies
1. Improper clutch adjustments	Need to correct it
2. The spring in the gear changing arrangement may not working	Need to change with a new spring
3. There may be sticking or jamming in the main shaft	Should correct the shaft
4. There may be a sticking of fork in the gear shifting system	Need to clean and lubricate.

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